# Used Car Price Analysis (Saudi Arabia)

العربية

## Overview

An end-to-end analysis of used car prices: data cleaning in **SQL Server**, then an interactive dashboard in **Power BI**.

## Business Problem

Buyers cannot tell whether a used car price is fair, and sellers do not know how to price their cars.

**Goal:** find which factors affect the price most, and give a clear takeaway for buyers and sellers.

## Questions

- How does price change with mileage?
- Which brands lose value fastest?
- Does the city affect the price?
- Does the gear type affect the price?

## Dataset

- **Synthetic dataset** generated for practice (not real listings)
- 2,520 rows, 12 columns: make, model, year, mileage, region, gear type, engine size, fuel type, origin, options, negotiable, price (SAR)
- Intentional data quality issues were included to practice cleaning

## Tools

- SQL Server (SSMS)
- Power BI

## Data Cleaning (SQL)

The raw table was never modified. All cleaning was written into a new table, `UsedCarClean`.

| Issue | Count | Decision |
|---|---|---|
| Duplicate rows | 20 | Removed with `ROW_NUMBER()` |
| Price = 0 (price not listed) | 297 (~12%) | Set to `NULL` with `NULLIF`, excluded from price analysis so averages are not distorted |
| Missing engine size | 40 (~1.6%) | Left as `NULL`, not used in the analysis |
| Missing mileage | 26 (~1%) | Left as `NULL`, excluded from mileage questions only |

## Key Findings

- **Mileage:** average price falls steadily, from about 145K SAR (under 50K km) to about 27K SAR (300K to 350K km).
- **Brand:** Mercedes is the most expensive on average (about 168K SAR), followed by Lexus (about 122K) and BMW (about 95K). Kia is the lowest (about 33K).
- **Gear type:** automatic cars average about 78K SAR versus about 51K SAR for manual.
- **City:** averages range from about 84K (Al-Khobar) to about 66K (Hail), so the city effect is modest compared with mileage and brand.

## Limitations

- The data is synthetic, so findings describe this dataset only, not the real market.
- Brand averages mix different models, so cheaper or more expensive models in a brand shift its average.
- The gear type gap is partly explained by luxury brands being almost all automatic.
- City differences may reflect the car mix in each city, not the city itself.
- The last mileage bin (350K km and above) has few cars, so it is not reliable.

## Files

- `saudi_used_cars.csv` : dataset
- `sql/` : data check and cleaning queries

- `used_cars_dashboard.pbix` : Power BI report
- `images/` : dashboard screenshot

## Author

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